Cape Peninsula
University of Technology

SUBJECT GUIDE FOR:	2026
COURSE CODE:	APT470S
NQF LEVEL:	7

APPLICATIONS DEVELOPMENT THEORY 4

TERM 3 RESEARCH PROJECT

TOPIC:

AI-Driven Model Engineering: *A Systematic Literature Review of Intelligent Model Generation, Transformation, Validation, and Software Synthesis*

Submission Deadline: *By Close of Business (CoB) on 27 August 2026*

1. Introduction

Background and Research Motivation: Model-Driven Engineering (MDE) places models at the centre of software development. Models represent different system perspectives, including context, interactions, structure, and behaviour. Conventional MDE typically transforms a Computation-Independent Model into a Platform-Independent Model, then into a Platform-Specific Model, and finally into executable software. Despite its potential to improve abstraction, reuse, productivity, and platform independence, traditional MDE remains constrained by specialist skill requirements, tool complexity, rigid transformation rules, model-code inconsistencies, and the difficulty of maintaining models as systems and requirements evolve. Artificial intelligence - including large language models, machine learning, deep learning, multimodal foundation models, knowledge graphs, neuro-symbolic methods, and intelligent agents - creates opportunities to automate and augment model-engineering activities. The emerging research challenge is no longer limited to generating individual UML diagrams; it concerns how AI can support the complete, continuous, and trustworthy engineering of software models.

Definition of AI-Driven Model Engineering: AI-Driven Model Engineering (AI-DME) is defined in this study as the systematic application of artificial intelligence to generate, interpret, complete, transform, validate, optimise, synchronise, explain, and evolve software models throughout the software-development lifecycle. AI-DME should not be positioned as a replacement for established modelling standards, transformation engines, formal constraints, or human expertise. Rather, it introduces an intelligent layer that augments conventional model-engineering processes while remaining subject to deterministic validation and accountable human oversight.

Research problem: Traditional MDE promises improved abstraction, reuse, platform independence, and automated code generation. However, its adoption is limited by the expertise required to produce correct and complete models, the rigidity of manually defined transformations, modelling-tool complexity, weak model-code synchronisation, and the difficulty of evolving models alongside rapidly changing systems. Although AI techniques are increasingly used to interpret requirements, generate UML and domain-specific models, recommend model elements, validate models, synthesise transformations, generate code, and reverse-engineer systems, the evidence remains fragmented across disconnected tasks and technologies. There is no sufficiently integrated understanding of where AI is applied across the model-engineering lifecycle, how AI-generated models are validated, whether they are semantically faithful to requirements, what level of autonomy has been achieved, and whether AI-driven approaches provide measurable advantages over conventional MDE. The absence of a consolidated taxonomy, maturity model, evaluation hierarchy, and evidence-based reference architecture makes it difficult to assess the reliability, reproducibility, industrial readiness, and governance requirements of AI-Driven Model Engineering.

Research Aim: To systematically identify, classify, and critically evaluate the application of artificial intelligence across the model-engineering lifecycle and to develop an evidence-based taxonomy, maturity model, reference architecture, and future research agenda for AI-Driven Model Engineering.

Traditional MDE and AI-Driven Model Engineering:

Traditional Model-Driven Engineering	AI-Driven Model Engineering
Models are created mainly by human modellers.	Models are collaboratively generated and refined by humans and AI.
Transformations depend primarily on predefined rules.	Transformations may be learned, inferred, generated, & adaptively refined.
Requirements are translated manually into models.	AI interprets natural-language and multimodal requirements.
Validation relies mainly on manually specified constraints.	AI detects anomalies while deterministic tools verify conformance and correctness.
Model-to-code generation follows fixed templates.	AI supports context-aware, platform-sensitive synthesis and repair.
Models are often updated manually.	AI assists continuous requirements-model-code synchronisation.
Runtime adaptation is limited.	Runtime evidence can drive model evolution and self-adaptation.

2. Research Questions

Main research question: How is artificial intelligence transforming the generation, analysis, validation, transformation, and evolution of software models across the model-engineering lifecycle?

Landscape and scope

RQ1: What publication trends, research communities, application domains, and contribution types characterise AI-DME research?

RQ2: Which AI technologies are used, including LLMs, machine learning, deep learning, multimodal models, knowledge graphs, reinforcement learning, neuro-symbolic AI, and intelligent agents?

Model-engineering lifecycle

RQ3: At which stages of the model-engineering lifecycle is AI applied?

RQ4: Which modelling languages and artefacts are supported, including UML, SysML, BPMN, domain-specific languages, architectural models, and metamodels?

RQ5: How is AI used to transform natural-language or multimodal requirements into structural, behavioural, interaction, and architectural models?

Intelligence and automation

RQ6: What roles does AI perform, such as generator, recommender, critic, validator, transformation synthesiser, repair mechanism, or autonomous agent?

RQ7: What levels of automation are achieved, from human assistance to semi-autonomous and continuous autonomous model engineering?

Quality and validation

RQ8: How are AI-generated models evaluated for syntactic validity, semantic fidelity, completeness, consistency, traceability, and standards conformance?

RQ9: Which deterministic tools, metamodels, constraints, simulators, model checkers, and human-review mechanisms validate AI outputs?

Transformation and implementation

RQ10: How does AI support model-to-model, model-to-text, model-to-code, and code-to-model transformations?

RQ11: To what extent do AI-driven approaches improve productivity, model quality, maintainability, code quality, and platform adaptability compared with traditional MDE?

Trustworthiness and adoption

RQ12: What risks arise from hallucination, nondeterminism, model drift, privacy exposure, intellectual-property concerns, and automation bias?

RQ13: What factors influence industrial adoption, interoperability, scalability, and organisational readiness?

RQ14: What evidence exists regarding reproducibility, generalisability, and deployment in real-world projects?

Research contribution

RQ15: What taxonomy, maturity model, evaluation hierarchy, and reference architecture can guide trustworthy AI-Driven Model Engineering?

RQ15: What conceptual framework will be suitable for this study?

3. Scope of the Review

The review should cover the complete lifecycle rather than restricting the analysis to automated UML generation.

AI-DME area	Representative activities
Requirements understanding	Extract actors, entities, constraints, relationships, behaviours, ambiguities, & quality defects.
Requirements-to-model generation	Produce use-case, class, sequence, activity, state, architecture, and domain models.
Model completion	Recommend missing classes, attributes, relationships, states, transitions, or constraints.
Model quality assurance	Detect incorrect, incomplete, inconsistent, ambiguous, or non-conforming models.
Model repair	Correct structural, syntactic, semantic, and cross-view defects.
Model transformation	Generate or refine model-to-model transformation rules.
Model-to-code synthesis	Produce executable code, tests, configurations, and deployment artefacts.
Code-to-model extraction	Recover architecture, UML views, and domain abstractions from repositories.
Model-code synchronisation	Maintain consistency among requirements, models, implementations, and tests.
Model evolution	Update models as requirements, technologies, and environments change.
Runtime modelling	Update models using telemetry, logs, execution traces, and digital twins.
Self-adaptive systems	Use runtime models and AI to recommend or execute software reconfiguration.
Collaborative modelling	Support communication & joint decision-making among stakeholders, analysts, & developers.

Expected Research Gaps to Investigate:

- Overemphasis on class diagrams rather than complete multi-view systems.
- Limited support for behavioural, architectural, and runtime models.

- Use of syntactic validity as a substitute for semantic correctness.
- Weak cross-diagram consistency and requirements traceability.
- Limited integration with metamodels, OCL, formal constraints, simulation, and model checking.
- Insufficient evidence on transformation correctness and model-code synchronisation.
- Reliance on small textbook examples and synthetic requirements.
- Lack of standard benchmarks and industrial-scale datasets.
- Inadequate comparison with expert modellers and traditional MDE tools.
- Weak reporting of prompts, model versions, stochastic variability, and cost.
- Limited uncertainty quantification, explainability, and governance.
- Insufficient evidence for safety-critical, regulated, multilingual, and domain-specific settings.
- Little support for continuous runtime model evolution and autonomous adaptation.

4. Multidimensional Taxonomy

A. Model-engineering stage

Requirements modelling, Domain modelling, Architectural modelling, Behavioural modelling, Model validation and repair, Model transformation, Code generation, Reverse engineering, Model evolution, Runtime adaptation.

B. Input artefact

Natural-language requirements, User stories and use cases, Existing diagrams, Source code, Database schemas, Documentation, Execution logs, Images and sketches, Metamodels, Transformation rules.

C. Output artefact

UML, SysML, and BPMN models; domain-specific models and metamodels; transformation scripts; formal constraints; source code and tests; architecture descriptions; runtime adaptation plans.

D. AI technology

Machine learning, Deep learning, Transformers and LLMs, Multimodal foundation models, Retrieval-augmented generation, Knowledge graphs and graphs neural networks, Reinforcement learning, Neuro-symbolic AI, Single-agent and multi-agent systems.

E. AI role

Requirement interpreter, Model generator or completer, Recommender, Critic and validator, Repair assistant, Transformation synthesiser, Code generator, Consistency monitor, Autonomous orchestrator.

F. Validation mechanism

Human expert review, Syntax parser and metamodel conformance, OCL and formal constraints, Consistency checking, Simulation and model checking, Transformation execution, Compilation and test execution, Requirements traceability, Hybrid deterministic-AI validation.

G. Quality attributes

Correctness, Completeness, Consistency, Understandability, Traceability, Maintainability, Semantic fidelity, Standards conformance, Transformation accuracy, Generated-code quality, Reliability and explainability.

AI-DME Automation Continuum:

Maturity level	Description
Level 1 - Manual modelling	No substantive AI support.
Level 2 - Assistive modelling	AI provides explanations, recommendations, and local improvements.
Level 3 - Interactive co-modelling	Human and AI iteratively construct and refine models.
Level 4 - Semi-autonomous modelling	AI generates substantial models or transformations requiring human approval.
Level 5 - Verifier-guided autonomous modelling	AI generates, validates, diagnoses, and repairs models using deterministic feedback.
Level 6 - Continuous adaptive modelling	AI agents maintain requirements-model-code-runtime alignment and support autonomous evolution.

5. Systematic Review Methodology

5.1 Review guidance and reporting

The review should follow established software-engineering SLR guidance and PRISMA reporting principles. A protocol should be developed and, where possible, registered or published before the full search and screening process begins.

5.2 Databases

- Scopus
- Web of Science
- IEEE Xplore
- ACM Digital Library
- ScienceDirect
- SpringerLink

Google Scholar may be used for supplementary backward and forward snowballing, but it should not serve as the sole primary database.

5.3 Proposed publication period

January 2015 to the final search date (2026).

5.4 Illustrative search string

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("artificial intelligence" OR "machine learning" OR "deep learning" OR
 "generative AI" OR "large language model*" OR LLM* OR
 "foundation model*" OR "multimodal model*" OR
 "intelligent agent*" OR "agentic AI" OR "knowledge graph*" OR
 "neuro-symbolic")
AND
("model-driven engineering" OR "model-driven development" OR
 "model-based software engineering" OR "software model*" OR
 UML OR SysML OR BPMN OR "domain-specific language*" OR metamodel*)
AND
("model generation" OR "model transformation" OR "model validation" OR
 "model completion" OR "model repair" OR "model evolution" OR
 "code generation" OR "reverse engineering" OR
 "requirements-to-model" OR "model-to-code")
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6. Inclusion and Exclusion Criteria

Include studies that	Exclude studies that
• Apply an identifiable AI technique to a model-engineering task.	• Discuss AI or MDE only conceptually without a technical contribution.
• Address software or systems modelling.	• Use “model” only to mean a predictive machine-learning model.
• Generate, analyse, validate, transform, repair, synchronise, or evolve models.	• Generate code directly without using or producing a software model.
• Present a framework, method, algorithm, tool, dataset, or empirical evaluation.	• Address non-software CAD models outside systems engineering.
• Provide sufficient technical and methodological information.	• Are editorials, abstracts, posters, or tutorials without full results.
• Are peer-reviewed and fall within the selected period.	• Duplicate a later or extended version of the same study.

Quality-Assessment Checklist: Each item may be scored as 0 = No, 0.5 = Partial, and 1 = Yes. Scores should inform evidence weighting rather than automatically excluding every lower-scoring study.

Criterion	Quality question
Research clarity	Are the aim and research problems clearly stated?
Context	Is the software and modelling context adequately described?
AI transparency	Are the AI model, version, configuration, prompts, and adaptation strategy reported?
Modelling transparency	Are the modelling language, metamodel, and artefacts identified?
Dataset quality	Are the dataset origin, composition, and preparation described?
Baseline	Is there an appropriate human, rule-based, or non-AI baseline?
Syntactic validity	Is grammar or metamodel conformance evaluated?
Semantic fidelity	Is alignment with source requirements or system meaning assessed?
Completeness	Are omitted entities, relationships, behaviours, or constraints measured?
Cross-view consistency	Are relationships among multiple model views evaluated?

Independent validation	Are outputs validated independently of the generating AI?
Human expertise	Are qualified modellers or domain experts involved where necessary?
Experimental rigour	Are repeated runs, uncertainty, and variability reported?
Statistical analysis	Are appropriate statistical tests or confidence intervals used?
Reproducibility	Are prompts, code, data, and generated models available?
Practical relevance	Is the approach evaluated on realistic or industrial systems?
Scalability	Is performance examined on large or complex systems?
Threats to validity	Are limitations and potential bias discussed?

7. Data-Extraction Framework

- Publication year, venue, country, and institution
- Study and contribution type
- Application domain and system scale
- AI technique, model, version, and openness
- Prompting, training, retrieval, or adaptation method
- Single-agent or multi-agent design
- Model-engineering stage and task
- Modelling language, metamodel, and diagram type
- Input and output artefacts
- Dataset and benchmark
- Baseline and comparison method
- Validation mechanism and human involvement
- Quality metrics and reported performance
- Computational cost and scalability
- Explainability and uncertainty treatment
- Standards conformance and traceability
- Reproducibility assets
- Reported benefits and limitations
- Industrial evidence and assigned maturity level

8. Evidence-Synthesis Strategy

Descriptive analysis

Publication trends; leading venues and countries; dominant AI technologies; supported modelling languages; model types; diagram types; application domains; contribution types; and industrial participation.

Thematic synthesis

Requirements-to-model generation; collaborative modelling; semantic quality; intelligent transformation; model-to-code and code-to-model engineering; verifier-guided generation; agentic workflows; model-code synchronisation; runtime models; and trustworthy automation.

Comparative synthesis

AI-generated versus human-generated models; AI-driven versus rule-based transformations; LLM-only versus verifier-guided methods; prompting versus fine-tuning; proprietary versus open models; single-agent versus multi-agent systems; textbook examples versus industrial projects.

Quantitative synthesis

A meta-analysis should be attempted only when tasks, datasets, baselines, and metrics are sufficiently homogeneous. Otherwise, structured narrative synthesis, effect-direction analysis, and carefully normalised comparisons are more defensible.

9. Anticipated Scholarly Contributions

- A comprehensive synthesis of AI applications across the model-engineering lifecycle.
- A precise definition and boundary of AI-Driven Model Engineering.
- A multidimensional taxonomy of technologies, roles, artefacts, tasks, and validation mechanisms.
- A seven-level model-quality and assurance hierarchy.

- An AI-DME maturity model from assistance to continuous autonomy.
- A verifier-guided reference architecture integrating AI, modelling tools, standards, and human oversight.
- A critical assessment of empirical quality, reproducibility, scalability, and industrial readiness.
- A catalogue of datasets, benchmarks, evaluation metrics, and tools.
- A prioritised research agenda for trustworthy and autonomous model engineering.
- An openly available evidence repository containing included studies, extraction data, and coding decisions.
- A conceptual framework suitable for the study.

10. Conclusion

This study should be positioned beyond automated diagram generation. Its principal novelty should lie in explaining how AI can support the complete and continuous engineering of models - from requirements interpretation to validated models, executable software, runtime feedback, and self-adaptation. The scientifically defensible position is not that AI can independently engineer trustworthy models, but that AI should operate within a verifier-guided, standards-compliant, traceable, and human-accountable environment.

Appendixes

A. Formatting Requirements

- **Font**: Times New Roman, 11 pt
- **Line Spacing**: 1.5
- **Margins**: 1 inch (2.54 cm) all around
- **Justification**: Fully justified
- **Length**: 20–30 pages (excluding references and appendices)
- **Headings**: Follow APA-style hierarchy
- **Figures/Tables**: Numbered with titles and sources (if applicable)
- **Referencing**: APA 7th Edition
- **Submission Format**: Word (docx)
- **Declaration**: Declare the type and level of AI usage

B. Added Notes:

- Use critical thinking – do not just summarize; you should compare, analyze the strengths and weaknesses of each paper, etc.
- Be aware of current AI-Driven Model Engineering.
- Compare multiple sources and identify patterns.
- Comparison table - Use tables to compare many things.
- Use PRISMA flowchart for selection of literature.
- Follow the PRISMA framework strictly and include the flow diagram.
- Taxonomy diagram - Include a detailed PRISMA diagram to show your review process and intersection of main concepts.
- Gap matrix - What has been covered vs what is lacking in current research (underexplored areas).
- Highlight any under-researched areas.
- Avoid plagiarism and AI-generated content.
- A lot of references and citations are required, even within tables.

C. Sample Study Selection Process

- **Initial Retrieval**: ~1,236 articles retrieved based on search criteria
- **Screening**: Titles and abstracts were reviewed to remove irrelevant works
- **Eligibility Check**: Full-text analysis for relevance to your research topic
- **Final Inclusion**: 86 papers included for qualitative synthesis

Database	Search Results	After Screening	After Full Review
IEEE Xplore	318	61	22
SpringerLink	211	37	18
ScienceDirect	252	45	13
ACM DL	178	29	10
PubMed	99	23	9
Scopus	140	41	14
Web of Science	38	6	0 (duplicates removed)
Total	**1,236**	**242**	**86**

D. Sample Table for Analysis

Bio-Inspired Algorithms based on ROAMS Objectives

Algorithm	Optimization Objectives	Use Cases	Strengths	Limitations
GA	Energy, latency, reliability	Multi-hop wearable routing	Good global search	Slow convergence
PSO	Power control, clustering	ECG streaming, fall detection	Fast, simple	Premature convergence
ACO	Routing, scheduling	Emergency routing	Adaptive path optimization	Slow in large-scale
ABC	Channel assignment	Mobile patient clustering	Balanced search	Suboptimal for large-scale
GWO	QoS-aware routing	Sleep/ICU monitors	Few parameters, strong	Still emerging
Hybrid (e.g., GA-PSO)	Multi-objective	ICU, disaster, mobile clinics	Robust, adaptive	Complex tuning

Performance Results Across Models

Algorithm	Latency	Energy Efficiency	PDR	Overhead	Best Use Case
GA	Medium	High	Medium	High	Long-term health tracking
PSO	Low	Medium	High	Medium	Real-time alerts
ACO	Medium	Medium	High	High	Reliable emergency comms
ABC	High	Medium	Medium	Low	Lightweight wearable ops
GWO	Medium	High	Medium	Low	Adaptive ICU networks
Hybrid GA-PSO	Low	High	High	High	Mobile, dense environments

E. Bloom's Taxonomy of Measurable Verbs

Benjamin Bloom created a taxonomy of measurable verbs to help us describe and classify observable knowledge, skills, attitudes, behaviors and abilities. The theory is based upon the idea that there are levels of observable actions that indicate something is happening in the brain (cognitive activity.) By creating learning objectives using measurable verbs, you indicate explicitly what the student must do in order to demonstrate learning.

Verbs that demonstrate *Critical Thinking*

					EVALUATION
					Appraise
				SYNTHESIS	Argue
				Arrange	Assess
			ANALYSIS	Assemble	Choose
			Analyze	Collect	Compare
		APPLICATION	Appraise	Combine	Conclude
		Apply	Categorize	Comply	Estimate
	COMPREHENSION	Complete	Compare	Compose	Evaluate
	Compare	Construct	Contrast	Construct	Interpret
KNOWLEDGE	Describe	Demonstrate	Debate	Create	Judge
List	Discuss	Dramatize	Diagram	Design	Justify
Name	Explain	Employ	Differentiate	Devise	Measure
Recall	Express	Illustrate	Distinguish	Formulate	Rate
Record	Identify	Interpret	Examine	Manage	Revise
Relate	Recognize	Operate	Experiment	Organize	Score
Repeat	Restate	Practice	Inspect	Plan	Select
State	Tell	Schedule	Inventory	Prepare	Support
Tell	Translate	Sketch	Question	Propose	Value
Underline		Use	Test	Setup	

F. Bloom's Taxonomy of Measurable Verbs

A sample article is attached for your reference. You may also search for similar research articles related to this research area.

G. PRISMA Framework and guidelines

Please visit [*https://www.prisma-statement.org/*] to download the PRISMA Framework and guidelines. The latest PRISMA framework is appropriate for this study.

H. ASSESSMENT RUBRIC

The assessment rubric is attached.